Statistical Review v2 - 2026-07-23 (Referee Report)

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Status: the headline finding is fixed in code; the remaining findings are documented as known limitations. This report is written in the register of a thesis committee referee. It supersedes nothing in the v1 review (`statistical-review-2026-07-23.md`); it corrects one thing the v1 review got wrong and records issues v1 did not examine.

Summary judgment

The project is methodologically serious for its scope: it separates a distribution-free guarantee (Tier 1) from a probability-weighted best-response (Tier 2 / Daily Double), it tests exact math by hand, and it labels its placeholders. That is above the bar for what it is. However, the Daily Double engine — the one genuinely statistical component — contained a second, subtler error beneath the one v1 fixed, and several modeling assumptions are stated too weakly. One of these (Finding 1) is a real defect that changed recommendations; it is now fixed. The rest are correctly scoped as limitations but must be stated plainly, which they now are.

Finding 1 (defect, FIXED): the comparison variance contradicted the simulation design

Context. The Daily Double engine estimates each candidate wager's win probability by Monte Carlo rollout, using **common random numbers** (CRN): within a trial, every candidate wager is scored against the *same* replayed rest-of-game randomness. CRN is a standard variance-reduction technique whose entire purpose is to make *comparisons between wagers* more precise by inducing positive correlation between their estimates.

The error. The v1 fix selected the recommended wager using a "plateau" of statistically-indistinguishable wagers, but computed the comparison standard error with the **independent** two-sample formula `sqrt(se_w^2 + se_best^2)`. That formula assumes the two estimates are independent. Under CRN they are not — they are strongly positively correlated. The correct variance of the difference is

```
Var(p_w - p_best) = Var(p_w) + Var(p_best) - 2*Cov(p_w, p_best),
```

and the `-2*Cov` term is exactly the variance reduction CRN was introduced to buy. By using the independent formula, the engine discarded that reduction and systematically *overstated* the comparison uncertainty.

Evidence. Instrumenting the engine's own CRN loop to record per-trial paired win indicators, for the exact reported scenario `scores = [4200, 6000, 3400]`, `p_correct = 0.42`, 3000 trials, seed 42:

quantity	value
independent comparison SE (used by v1)	~0.0114, roughly constant across wagers
true paired comparison SE	falls from 0.0095 to 0.000 as wagers approach the best
win-indicator correlation with the best wager	rises from ~0.30 to 1.00

The decision consequence was not cosmetic. Under the independent SE the plateau spanned **$100–$4,200 and the engine recommended $100**; under the correct paired SE the plateau was **{$4,200}**. The engine's stability across confidence levels — the property v1 was proud of — turned out to be an artifact of an over-wide (wrong) interval, and it caused systematic *under-betting*.

The fix. Two coupled changes:

1. *Paired comparison variance.* The engine now records the per-trial win indicator for every candidate wager and estimates the SE of the difference (candidate vs. empirical best) directly from the per-trial paired differences. This respects the CRN design.
2. *Multiplicity control.* Roughly `k` candidate wagers are each compared against the single empirical best wager. This is a multiple-comparisons-with-the-best problem (Hsu, 1996). A per-comparison `z = 1` re-introduces instability, because the empirical best is itself the maximum of `k` noisy estimates. The engine now uses a Bonferroni band `z* = Phi^{-1}(1 - alpha/k)` (default `alpha = 0.05`), which adapts to the number of candidate wagers (`z* ~ 2.7-3.05` for the grids used here).

Why the fix is correct on both ends (verified).

- *Flat scenario* (`[4200, 6000, 3400]`, `p ~ 0.43`), where all wagers are genuinely about equal: recommends **$0, stable across every seed** — the right call, made for the right reason (a wide plateau because the surface is truly flat, not because the interval is mis-computed).
- *Clear-signal scenario* (`[3000, 4000, 2000]`, guaranteed correct), where betting genuinely helps: recommends **$1,200, stable across every seed**, and does **not** collapse to $0 — the paired SE detects that wagers below $1,200 are significantly worse, and the "smallest wager on the plateau" rule then picks $1,200 rather than the full $3,000.

This is the difference between a rule that is stable because it is right and a rule that is stable because it is blunt.

Finding 2 (limitation, DOCUMENTED): the reported win probability is selection-biased

`win_probability` is the in-sample maximum over ~`k` candidate wagers. The maximum of `k` noisy estimates is an upward-biased estimate of the true best value (the "winner's curse"). The *recommended wager* is well insulated — it is the smallest wager on a multiplicity-controlled plateau, not the raw argmax — but the reported *percentage* should be read as a mild optimistic upper bound, not an unbiased point estimate. This is now stated in the engine docstring, the notes returned to the caller, and the risk register. A future version could report a debiased estimate (e.g., data-splitting: choose the wager on one half, estimate its win probability on the other) or the recommended wager's own out-of-selection estimate.

Finding 3 (limitation, DOCUMENTED): "monotonicity" was overstated

The prior docstring implied CRN delivered monotonicity guarantees. It does not. Because opponents' modeled Final Jeopardy wagers depend on the player's own score, raising the Daily Double wager can change opponents' downstream behavior and move the win probability non-monotonically. CRN provides a fair *paired comparison*, not monotonicity. Wording corrected in the engine and the white paper.

Finding 4 (limitation, DOCUMENTED): Tier 2 opponent-correctness independence

The exact Final Jeopardy enumeration treats each opponent's correctness as independent. Real Final Jeopardy correctness is positively correlated across players (shared clue, shared difficulty). Positive correlation would, for example, raise the probability that *both* leaders miss — so the "third place can win when both leaders bust" figure (1/12 in the shipped example) is, if anything, a conservative floor under a more realistic

positively-correlated model. This is a modeling choice, not an arithmetic error, and is now flagged against ADR-0002 and in the risk register.

Finding 5 (limitation, DOCUMENTED): "exact" means double precision, not exact rational arithmetic

The Tier 2 enumeration is called "exact" to contrast it with Monte Carlo. It is exact in the sense of *no sampling error*, but it is computed in IEEE-754 double precision, not exact rational arithmetic. The accumulated floating error (~1e-16) is far below the 1e-9 comparison tolerance, so no result is affected, but the white paper wording is tightened to "exact enumeration in double precision (no sampling error)".

Finding 6 (limitation, DOCUMENTED): the app and the reference engine use different PRNGs

The Python engine uses the Mersenne Twister; the JavaScript app uses `mulberry32`. Given the "same" seed they produce *different* random streams, so the app's Daily Double numbers will not match the Python engine's trial-for-trial. The exact (Final Jeopardy) functions and the deterministic bound functions do match, and were re-verified under Node against the Python hand-derived values, along with the inverse-normal and Bonferroni-band values and the paired-SE <= independent-SE invariant. But the Monte Carlo Daily Double outputs are only distributionally comparable, not identical. Extends risk #8.

Finding 7 (limitation, DOCUMENTED): the board abstraction is coarse

`approximateClueList` reconstructs the remaining board from a single number (total dollars left) by assuming a fixed average clue value. Two very different board states (many small clues vs. a few large ones) map to the same rollout. Already labeled a placeholder; reinforced here because it materially affects Daily Double rollouts far from Final Jeopardy.

What would move this from "good project" to "defensible thesis chapter"

1. Calibrate the opponent model against real data (the single largest source of unquantified error; ADR-0002, risk #5).
2. Replace the Bonferroni band with the exact MCB constant (Hsu) or a bootstrap over the joint rollout, and report a proper simultaneous confidence set for the win-probability curve.
3. Debias the reported win probability (data-splitting), and show the coverage of the plateau empirically.
4. Model opponent-correctness correlation and report sensitivity of the headline numbers to it.
5. Unify the two engines behind one PRNG (or generate the JS from the Python) so parity is exact and CI-checkable.

None of these are required for the tool to be useful. All of them are the difference between "the numbers are directionally sound and honestly labeled" and "the numbers are defended".

Validation performed for this review

- Instrumented CRN experiment demonstrating the independent-vs-paired SE gap and the resulting opposite recommendations.
- Cross-seed stability of the corrected (paired SE + Bonferroni) rule on both a flat scenario and a clear-signal scenario.
- Full test suite: 33 passing (4 new/updated tests for the paired SE, the Bonferroni band, and clear-signal behavior).
- JS/Python re-verification under Node: exact FJ values, inverse-normal and Bonferroni-band reference values, and the paired-SE <= independent-SE invariant all match.